Review article: Numerical Methods for Engineering Quantum Computers

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ABSTRACT

In the article by Herrmann et al, "Quantum utility – definition and assessment of a practical quantum advantage," the quantum computer "Application Readiness Level" (ARL) is defined: (ARL1) "concept with unknown potential," (ARL2) "beneficial in small idealized systems," (ARL3) "utility indicated by theory or resource estimations," (ARL4) "simulated utility demonstration" and (ARL5) "utility demonstration on quantum hardware." Hermann et al posit that ARL3 has been reached via demonstrations of the Variational Quantum Eigenvalue (VQE) algorithm. In this review article, the latest activities from the computational "digital twin" perspective are reported on in which scientists are working hard to bring practitioners out of the Noisy Intermediate Scale Quantum (NISQ) era into an era of "Quantum supremacy." In other words, this review article discusses digital twin activities that help to advance from ARL3 to ARL4, or even ARL5. The three main categories in this review article are (1) Optimization, Control, Inverse problems, and exploratory simulation algorithms for Schrodinger's equation, (2) Quantum computer Error correction, and (3) Quantum algorithms for benchmarking quantum computers (to include emulator algorithms like Qiskit).

1. Introduction

The motivation for this review article is the fact that Moore's law is no longer applicable [93]. As quoted from Shalf [113], "Moore's Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the ×86, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore's Law for the past 50 years is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore's

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Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling."

References one through three in the above quote correspond to the following references respectively: [93], [88], [90].

As further motivation for this review article, in Figure 3 of the article by Shalf [113], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three categories: (a) roadmap for the next ten years, (b) 20 years, and (c) "Decades beyond exascale," "New Models of Computation." For category (c), Shalf [113] refers to the following prospective technology: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorhic, and (v) quantum [35, 36].

Of the "new models of computation" listed in Shalf's article [113], this review article will address the "Numerical Methods for Engineering Quantum Computers" aspects associated with the emerging "quantum computing" paradigm. As outlined by Gamble [39], the development of reliable quantum computers will have a tranformative effect on computer technology. A perfectly working fifty qubit quantum computer can efficiently process 2^{50} pieces of data which is 1024 Terabytes (1 Petabyte)!

That being said, right now we are in the "Noisy Intermediate-Scale Quantum" [20] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

2. Optimization, Control, Inverse problems, and exploratory simulation algorithms for Schrodinger's equation

In 1982, Feit et al [33] developed a spectral method for determining the eigenvalues and eigenfunctions of the time independent Schrodinger equation:

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2M}\nabla^2\psi + V(x,y)\psi \tag{1}$$

$$\psi(x,y,t) = \sum_{n,j} A_{nj}u_{nj}(x,y)e^{-iE_n t} \tag{2}$$

$$E_n u_{nj} = -\frac{1}{2M}\nabla^2 u_{nj} + V u_{nj}, \tag{3}$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}. \tag{4}$$

In 2002, Bao et al [6] developed a time splitting spectral approximation for the time dependent Schrodinger equation in the "semiclassical" regime:

$$iu_t^\epsilon = -\frac{\epsilon}{2}\Delta u^\epsilon + \frac{1}{\epsilon}V(x)u^\epsilon \quad x \in R^d \quad u^\epsilon(x,0) = u_0^\epsilon(x) \tag{5}$$

Bao et al report results in one and two space dimensions.
In 2004, Zheng et al [131] developed a finite element method for finding the ground state of the lithium atom. In otherwords, Zheng et al found the smallest eigenvalue and the associated eigenfunction for the time independent Schrodinger equation:

$$E\psi = H\psi \tag{6}$$

where the Hamiltonian operator H is defined as

$$H = -\frac{1}{2}(\Delta_1 + \Delta_2 + \Delta_3) - \frac{3}{r_1} - \frac{3}{r_2} - \frac{3}{r_3} + \frac{1}{r_{12}} + \frac{1}{r_{13}} + \frac{1}{r_{23}} \tag{7}$$

$$\Delta_j = \frac{\partial^2}{\partial x_j^2} + \frac{\partial^2}{\partial y_j^2} + \frac{\partial^2}{\partial z_j^2} \quad 1 \le j \le 3 \tag{8}$$

$$r_{jk} = |\vec{r}_j - \vec{r}_k| \quad r_j = |\vec{r}_j| \tag{9}$$

After converting from cartesian coordinate system to spherical coordinate system, the authors were able to reduce the dimension of the problem from 9 to 6.

In Grivopoulos' 2005 Phd thesis [43], methods were developed for the control of quantum systems. We highlight one such control problem discussed in the thesis. The Lindblad equation [89, 81] for open quantum systems is (see 3.7 in [43]):

$$\dot{\rho} = -i[H, \rho] + \sum_n K_n \rho K_n' - \frac{1}{2}\{\sum_n K_n' K_n, \rho\} \tag{10}$$

The Lindblad equation models the interaction of two quantum systems, A and B, with respective Hilbert spaces H_A and H_B. The Hilbert space for the composite space is,

$$H_{AB} = H_A \otimes H_B \tag{11}$$

A general state vector on the composite space is,

$$\psi_{AB} = \psi_A \otimes \psi_B. \tag{12}$$

As stated in [43], "For the open system A (interacting with its environment B), ρ is the state variable analogous to the vector ψ for an isolated quantum system." Expressions for ρ are:

$$\rho = \mathrm{tr}_B \psi_{AB} \psi_{AB}' = \sum_{ij} (\sum_I (\psi_{AB})_{iI}^* (\psi_{AB})_{jI}) e_j e_i' \tag{13}$$

Note that ρ is self-adjoint, positive semi-definite, and $\mathrm{tr}\rho = 1$. As an example of control of open quantum systems, [43], considers a control problem for the laser cooling of atoms and molecules. The Lindblad equation with Hamiltonian control is

$$\dot{\rho} = -i[H_0 + V u(t), \rho] + L(\rho), \tag{14}$$

where

$$L(\rho) = \sum_{ij} K_{ij} \rho K_{ij}^\dagger - \frac{1}{2}\{K_{ij}^\dagger K_{ij}, \rho\} \tag{15}$$

[43], for the 3×3 case, determines the control, $u(t)$, in (14), that maximizes $\rho_{11}(T)$ given $\rho(0) = \rho_0$.

In the article by Han et al [48], a neural network algorithm was devised for computing the ground state to the time independent Schrodinger equation (6). The Hamiltonian operator corresponds to the Born-Oppenheimer approximation [16] with N electrons and M ions:

$$\vec{r} = (\vec{r}_1, \dots, \vec{r}_N) \qquad \vec{R} = (\vec{R}_1, \dots, \vec{R}_M) \tag{16}$$

$$H(\vec{r}, \vec{R}) = -\frac{1}{2}\sum_{i=1}^{N} \nabla_i^2 + \sum_{i=1}^{N}\sum_{j=i+1}^{N} \frac{1}{|\vec{r}_i - \vec{r}_j|} - \sum_{I=1}^{M}\sum_{i=1}^{N} \frac{Z_I}{|\vec{r}_i - \vec{R}_I|} + \sum_{I=1}^{M}\sum_{J=I+1}^{M} \frac{Z_I Z_J}{|\vec{R}_I - \vec{R}_J|} \tag{17}$$

It is assumed that the first $N_\uparrow$ electrons are spin-up and the remaining $N - N_\uparrow$ electrons are spin down[37]. The trial wave functions are assumed in [48] to be real valued, anti-symmetric, and have the following form:

$$\psi(\vec{r}, \vec{R}) = S(\vec{r}, \vec{R}) A^\uparrow(\vec{r}^\uparrow) A^\downarrow(\vec{r}^\downarrow) \tag{18}$$

[48] represent S,$A^\uparrow$, $A^\downarrow$ using a deep neural network. The weights and biases of the network are trained in order to minimize:

$$E[\psi] = \frac{\psi^*(\vec{r}, \vec{R}) H \psi(\vec{r}, \vec{R}) d\vec{r}}{\psi^*(\vec{r}, \vec{R}) \psi(\vec{r}, \vec{R}) d\vec{r}}. \tag{19}$$

[48] represent the network parameters at epoch t as,

$$\vec{\Theta}, \tag{20}$$

and the cost function in terms of $T\vec{heta}$ is given as,

$$E[\psi] \equiv \Omega_{E_{ref}}(T\vec{heta}). \tag{21}$$

The stochastic gradient descent method with learning rate η is used to minimize (21):

$$\Theta_{t+1} = \Theta_t - \eta \nabla_{\vec{\Theta}} \Omega_{E_{ref}}(T\vec{heta}) \tag{22}$$

[48] report results within two percent error for H_2 (two electrons), Be_4 (four electrons), LiH_4 (four electrons), He (two electrons), and H_{10} (ten electrons).
[52] developed a similar method as [48] for computing the ground state to the time independent Schrodinger equation (6). Both [52] and [48] express the trial wave functions using a neural network:

$$\psi(\vec{r}, \vec{R}) \approx \psi_{\vec{\Theta}}(\vec{r}) \tag{23}$$

[52] report results within three percent error for H_2 (two electrons), Be_4 (four electrons), LiH_4 (four electrons), and B (five electrons).

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3. Quantum algorithms and Quantum emulators for benchmarking quantum computers

3.1. Quantum Emulators

The development of Quantum Emulators is essential for benchmarking quantum computers since (i) the level of noise in Quantum Emulators can be prescribed exactly thereby enabling practitioners to quantify the effect of noise on a given quantum circuit design, (ii) one can prescribe a given noise model[83] and then compare emulator results with an actual quantum computer in order to characterise the actual noise present in a quantum computer, and (iii) one can isolate quantum algorithmic problems from intrinsic quantum computer noise problems.

A Quantum algorithm consists of (i) state preparation, (ii) state measurement(s), and (iii) a sequence of (basic) unitary operation(s).

3.1.1. state preparation

Given n qubits, there will be $N = 2^n$ basis states. For example if $n = 3$, the standard basis is

$$\begin{aligned} |000\rangle &= |0\rangle \otimes |0\rangle \otimes |0\rangle \equiv \vec{\mathbf{0}} \\ |001\rangle &= |0\rangle \otimes |0\rangle \otimes |1\rangle \equiv \vec{\mathbf{1}} \\ &\dots \\ |111\rangle &= |1\rangle \otimes |1\rangle \otimes |1\rangle \equiv \vec{\mathbf{7}} \end{aligned}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In general if one wants to prepare the state,

$$|\vec{\psi}\rangle \qquad |\vec{\psi}\rangle \in C^N,$$

on a quantum computer, then one searches for as sparse a unitary matrix U_ψ as possible such that,

$$U_\psi |\vec{\mathbf{0}}\rangle = |\vec{\psi}\rangle,$$

or, equivalently,

$$U_\psi |\vec{\mathbf{0}}\rangle = \frac{e^{i\theta}}{||\psi||} \sum_{i_1 i_2 \dots i_n} \psi_{i_1 i_2 \dots i_n} |i_1 i_2 \dots i_n\rangle .$$

The indices i_k take values in 0, 1, $\theta \in [0, 2\pi)$, and

$$||\psi||^2 = \sum_{i_1 i_2 \dots i_n} |\psi_{i_1 i_2 \dots i_n}|^2$$

In preparing an n qubit quantum state, if no ancillary qubits are to be used, then the circuit depth is $O(2^n)$ where the circuit depth is the number of basic quantum computer operations that cannot be performed in parallel. As reported in [59, 102], the lower and upper bounds on the number of CNOT gates required for state preparation of n qubits, with no additional ancillaries, is $(1/2)2^n$ and $(23/24)2^n$ respectively. As reported in [130], an optimal circuit depth of $O(n)$ can be achieved if the number of ancilla qubits is $O(2^n)$ [119],[109] (Corollary 1.9).

The results in [119, 109, 102] describe a very important trade off. The larger the state preparation circuit depth, the higher the odds that deocoherence ruins the prepared state, on the other hand, the larger the number of ancillary qubits, the larger the quantum computer that must be built. The quantum singular value decomposition of a dense matrix[110] is an example in which the trade-off between circuit depth and number of ancillar qubits must be considered.

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- Parenthetical: `\citep{WB96}` produces (Wettig & Brown, 1996).
- Textual: `\citet{ESG96}` produces Elson et al. (1996).
- An affix and part of a reference: `\citep[e.g.][Ch. 2]{Gea97}` produces (e.g. Governato et al., 1997, Ch. 2).

In the numbered scheme of citation, `\cite{<label>}` is used, since `\citep` or `\citet` has no relevance in the numbered scheme. natbib package is loaded by cas-sc with `numbers` as default option. You can change this to author-year or harvard scheme by adding option `authoryear` in the class loading command. If you want to use more options of the natbib package, you can do so with the `\biboptions` command. For details of various options of the natbib package, please take a look at the natbib documentation, which is part of any standard LaTeX installation.

A. My Appendix

Appendix sections are coded under `\appendix`.

`\printcredits` command is used after appendix sections to list author credit taxonomy contribution roles tagged using `\credit` in frontmatter.

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